

Problem 8.16

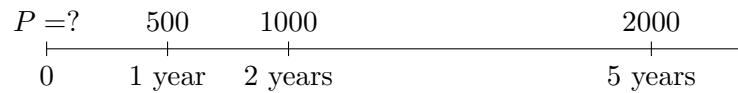
The effective rate of discount is 5%. Linda will receive \$500 one year from today, \$1,000 two years from today, and \$2,000 five years from today. Find the combined present value of her three future cash flows.

Solution

The effective discount rate is

$$d = 5\% = 0.05.$$

We represent the cash flows on a timeline:



Using the effective discount rate, the present value of a payment received at time t is multiplied by $(1 - d)^t$. Therefore,

$$P = 500(1 - d)^1 + 1000(1 - d)^2 + 2000(1 - d)^5.$$

Substituting $d = 0.05$ gives

$$\begin{aligned} P &= 500 \left(1 - \frac{5}{100}\right)^1 + 1000 \left(1 - \frac{5}{100}\right)^2 + 2000 \left(1 - \frac{5}{100}\right)^5 \\ &= 500(0.95) + 1000(0.95)^2 + 2000(0.95)^5 \\ &\approx 2925.061875. \end{aligned}$$

Thus, the combined present value is

$$\boxed{P \approx \$2,925.06}.$$